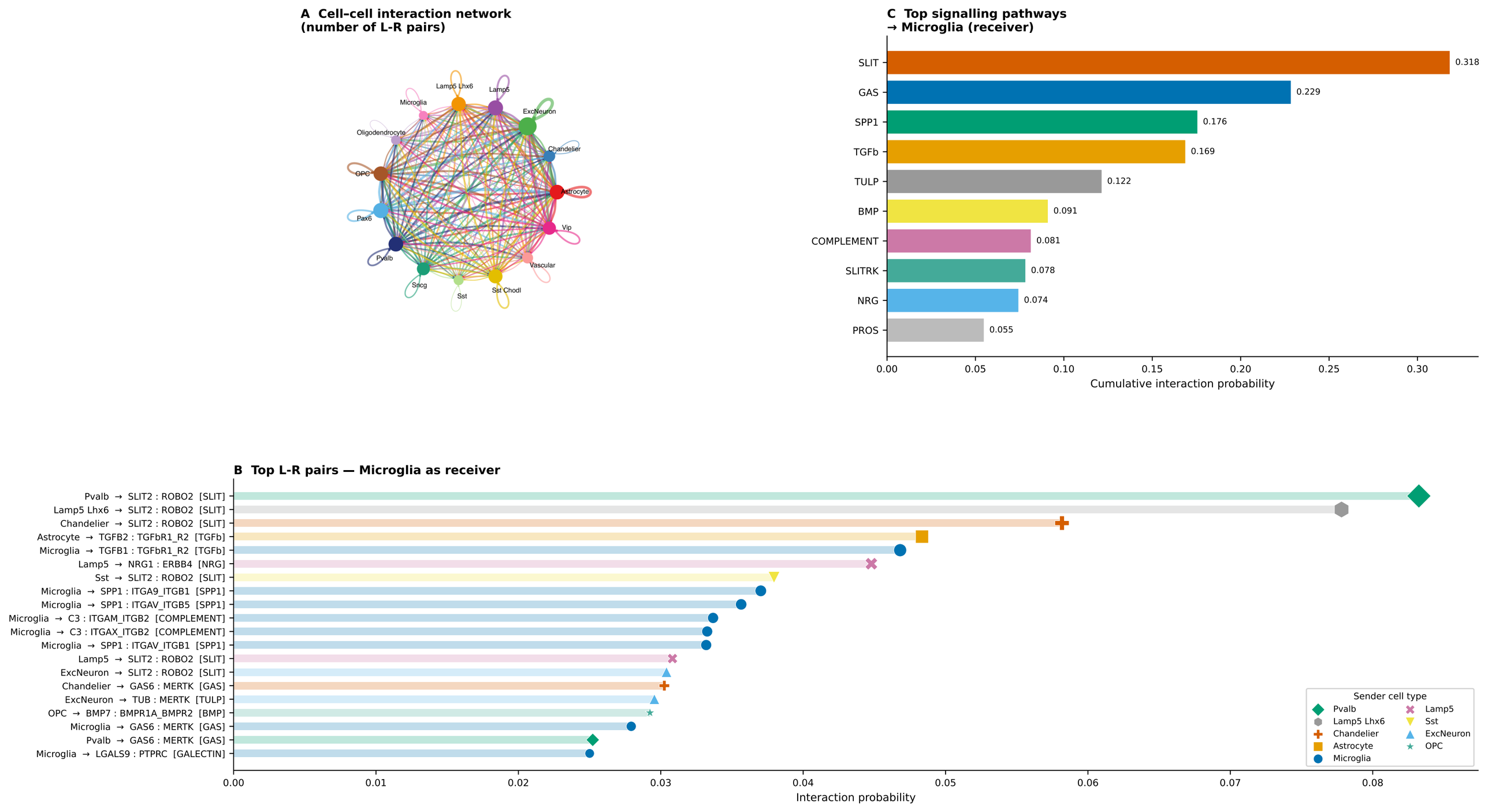
A Cell–cell interaction network
(number of L-R pairs)
Lamp5 Lhx6
Lamp5
Microglia
ExcNeuron
Oligodendrocyte
Chandelier
OPC
Astrocyte
Pax6
Vip
Pvalb
Vascular
Sncg
Sst
Sst Chodl
C Top signalling pathways
→ Microglia (receiver)
SLIT 0.318
GAS 0.229
SPP1 0.176
TGFb 0.169
TULP 0.122
BMP 0.091
COMPLEMENT 0.081
SLITRK 0.078
NRG 0.074
PROS 0.055
0.00 0.05 0.10 0.15 0.20 0.25 0.30
Cumulative interaction probability
B Top L-R pairs — Microglia as receiver
Pvalb → SLIT2 : ROBO2 [SLIT]
Lamp5 Lhx6 → SLIT2 : ROBO2 [SLIT]
Chandelier → SLIT2 : ROBO2 [SLIT]
Astrocyte → TGFB2 : TGFbR1_R2 [TGFb]
Microglia → TGFB1 : TGFbR1_R2 [TGFb]
Lamp5 → NRG1 : ERBB4 [NRG]
Sst → SLIT2 : ROBO2 [SLIT]
Microglia → SPP1 : ITGA9_ITGB1 [SPP1]
Microglia → SPP1 : ITGAV_ITGB5 [SPP1]
Microglia → C3 : ITGAM_ITGB2 [COMPLEMENT]
Microglia → C3 : ITGAX_ITGB2 [COMPLEMENT]
Microglia → SPP1 : ITGAV_ITGB1 [SPP1]
Lamp5 → SLIT2 : ROBO2 [SLIT]
ExcNeuron → SLIT2 : ROBO2 [SLIT]
Chandelier → GAS6 : MERTK [GAS]
ExcNeuron → TUB : MERTK [TULP]
OPC → BMP7 : BMPR1A_BMPR2 [BMP]
Microglia → GAS6 : MERTK [GAS]
Pvalb → GAS6 : MERTK [GAS]
Microglia → LGALS9 : PTPRC [GALECTIN]
0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08
Interaction probability
Sender cell type
Pvalb
Lamp5
Lamp5 Lhx6
Sst
Chandelier
ExcNeuron
Astrocyte
OPC
Microglia